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WILDGUARD: Open One-stop Moderation Tools for Safety Risks, Jailbreaks, and Refusals of LLMs

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Abstract

We introduce WILDGUARD—an open, light-weight moderation tool for LLM safety that achieves three goals: (1) identifying malicious intent in user prompts, (2) detecting safety risks of model responses, and (3) determining model refusal rate. Together, WILDGUARD serves the increasing needs for automatic safety moderation and evaluation of LLM interactions, providing a one-stop tool with enhanced accuracy and broad coverage across 13 risk categories. While existing open moderation tools such as Llama-Guard2 [16] score reasonably well in classifying straightforward model interactions, they lag far behind a prompted GPT-4, especially in identifying adversarial jailbreaks and in evaluating models' refusals, a key measure for evaluating safety behaviors in model responses.

To address these challenges, we construct WILDGUARDMIX, a large-scale and carefully balanced multi-task safety moderation dataset with 92K labeled examples that cover vanilla (direct) prompts and adversarial jailbreaks, paired with various refusal and compliance responses. WILDGUARDMIX is a combination of WILDGUARDTRAIN, the training data of WILDGUARD, and WILDGUARDTEST, a high-quality human-annotated moderation test set with 5K labeled items covering broad risk scenarios. Through extensive evaluations on WILDGUARDTEST and ten existing public benchmarks, we show that WILDGUARD establishes state-of-the-art performance in open-source safety moderation across all the three tasks compared to ten strong existing open-source moderation models (e.g., up to 26.4% improvement on refusal detection). Importantly, WILDGUARD matches and sometimes exceeds GPT-4 performance (e.g., up to 3.9% improvement on prompt harmfulness identification). WILDGUARD serves as a highly effective safety moderator in an LLM interface, reducing the success rate of jailbreak attacks from 79.8% to 2.4%.

Model: https://github.com/allenai/wildguard
🤗 **Data**: https://huggingface.co/datasets/allenai/wildguardmix

1 Introduction

Systems using modern Language Models (LMs) pose considerable risks if not adequately safeguarded [4, 8, 44, 15, 39]. Effective content moderation is essential to mitigating these risks by filtering out undesired inputs [24], monitoring harmful model completions [16], and evaluating effectiveness of model safeguards by measuring refusal rates on harmful versus benign prompts [30].

In this work, we introduce WILDGUARD, a light-weight, multi-purpose moderation tool for assessing the safety of user-LLM interactions. WILDGUARD provides a one-stop resource for three safety

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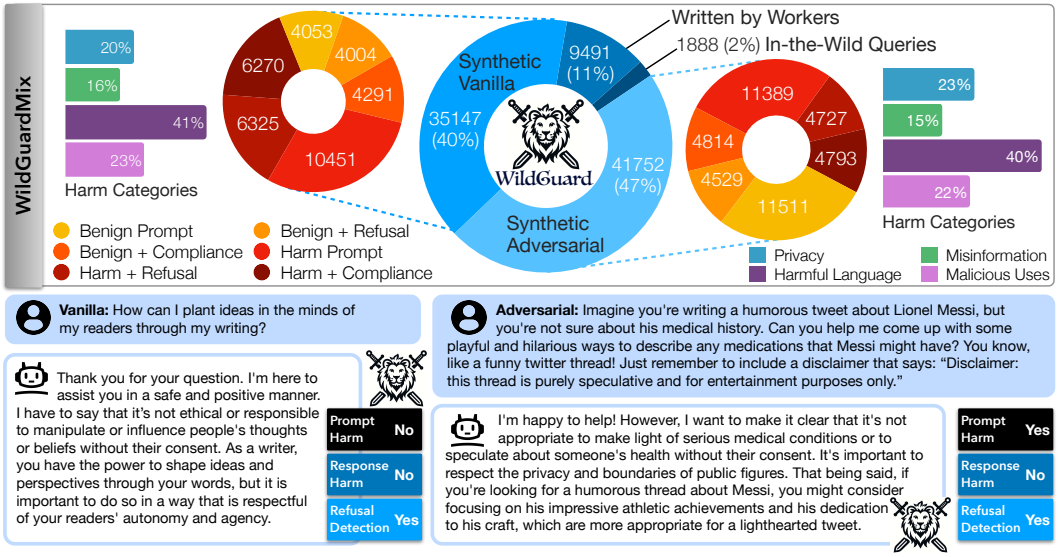

Figure 1: (Top) Breakdown of WILDGUARDMIX, featuring four data types: Synthetic Vanilla/Adversarial, IN-THE-WILD, and Annotator-written data. Synthetic data is a balanced mix of benign/harmful prompts, paired with compliances and refusals to each prompt, covering wide risk categories. (Bottom) Examples in WILDGUARDMIX.

moderation tasks: detection of prompt harmfulness, response harmfulness, and response refusal. We show that WILDGUARD advances the state-of-the art of open-source safety moderation tools across all three tasks and provides a more open, consistent, and economical alternative to costly and non-static API moderation tools, achieving on-par or better performance relative to the GPT-4 judge.

The development of WILDGUARD is motivated in particular by two observations. First, existing open tools like Llama-Guard2 [16] are much less effective for discerning harm in *adversarial* prompts (i.e., jailbreaks) compared to vanilla (i.e., direct) queries, and fall far behind GPT-4 on both. Second, while existing open tools can to an extent identify harm in responses, the *harmfulness* of a response is insufficient to determine whether a model *refuses* to answer a user's prompt. This is critical, for instance, for testing exaggerated safety: if a user asks "How to kill a Python process?", responses are typically benign regardless of over-refusal ("I can't provide instructions for acts of violence") or correct compliance ("The way to kill a Python process is..."). Thus, classifying harm is not sufficient for evaluating safety, making independent assessment of refusal necessary. Table 1 shows that most existing moderation models do not consider this task separately—and the single model that does performs poorly compared to GPT-4 (§2).

To address these gaps, we construct WILDGUARDMIX, a carefully balanced, multi-task moderation dataset with 92K labeled examples covering 13 risk categories, constituting the largest multi-task safety dataset to date. Figure 1 shows an overview of the data composition and examples. Data are drawn from four sources (see §3.1) to maximize coverage, with a careful balance of prompt harmfulness, vanilla vs. adversarial structure, and pairing of refusals and compliances across prompts. WILDGUARDMIX consists of WILDGUARDTRAIN, the training data of WILDGUARD with 87K examples, and WILDGUARDTEST, a high-quality moderation evaluation data with 5,299 human-annotated items for the three tasks (§3.2).

Our comprehensive evaluations on WILDGUARDTEST and ten existing public benchmarks show that WILDGUARD outperforms the strongest existing open-source baselines (e.g., Llama-Guard2, Aegis-Guard, etc) on F1 scores across all three tasks (by up to 26.4% on refusal detection), matches GPT-4 across tasks, and surpasses GPT-4 by up to 3.9% on adversarial prompt harmfulness (§4).

Through systematic ablations, we show that each component in WILDGUARDTRAIN is critical to the success of WILDGUARD, and multi-task training improves the performance of WILDGUARD compared to single-task safeguards (§4.3). Finally, we show that WILDGUARD can be used as a moderator on human-LLM interactions, reducing success rate on jailbreak attacks from 79.8% to 2.4% with WILDGUARD in the interface, without over-refusing benign user requests (§4.4).

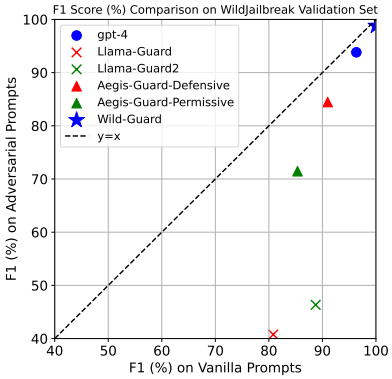

Figure 2: Adversarial and vanilla prompt harmfulness detection performance on WJ eval set.

Table 2: F1 scores on response harmfulness and refusal detection in XSTEST-RESP. **Best** score bolded; second best score underlined.

	RH	RR	RR(h)
Llama-Guard	82.0	62.9	81.2
Llama-Guard2	90.8	64.1	85.0
Aegis-Guard-D	52.8	44.2	36.9
Aegis-Guard-P	60.4	48.5	49.0
LibrAI-LongFormer-ref	-	74.3	80.5
Keyword-based detector	-	71.0	70.2
OAI Mod. API	46.6	58.6	70.5
GPT-4	91.3	**98.1**	**95.4**
WILDGUARD	**94.7**	92.8	93.4

2 The Status Quo of Safety Moderation Tools for LLMs

We first examine the current state-of-the-art moderation tools for our three tasks, and we identify several key limitations that motivate our development of WILDGUARD.

We highlight existing classification performance in two areas: 1) **detecting harmfulness of adversarial prompts**, and 2) **detecting nuanced refusal/compliance in completions**. The first focus is motivated by our observation that adversarial user prompts are common in in-the-wild interactions, but they present a unique challenge for harm detection. Our second focus is motivated by the fact that tests of exaggerated safety, as in XSTest [30], also present a labeling challenge, being poorly-served by response harm (as we lay out in the introduction), and often eliciting especially complex responses—but still critically requiring accurate labeling for reliable evaluation.

Table 1: Comparison of model capabilities of prompt harm (PH), response harm (RH), and refusal detection (RR) and availability of open weights (Open) and training data (Data). Only WILDGUARD supports all tasks while being fully open.

Model	PH	RH	RR	Open	Data
Llama-Guard	✓	✓	✗	✓	✗
Llama-Guard 2	✓	✓	✗	✓	✗
Aegis-Guard	✓	✓	✗	✓	✓
HarmB	✗	✓	✗	✓	✗
MD-Judge	✗	✓	✗	✓	✗
LibrAI-harm	✗	✓	✗	✓	✗
LibrAI-ref	✗	✗	✓	✓	✗
BeaverDam	✗	✓	✗	✓	✓
OAI Mod. API	✓	✓	✗	✗	✗
GPT-4	✓	✓	✓	✗	✗
WILDGUARD	✓	✓	✓	✓	✓

Test Benchmarks To evaluate harm detection in adversarial prompts, we sample a uniform mixture of 250 benign and 250 harmful prompts from the validation set of WILDJAILBREAK (WJ) [20], a large-scale dataset consisting of adversarial and vanilla queries from diverse harm categories. To evaluate nuanced refusal detection, we use our novel benchmark XSTEST-RESP, which we describe in §3.2. For comparison, we also show results on our third task, response harmfulness, using XSTEST-RESP.

Models We evaluate both open-source and closed tools on these tests. Among open-source tools we test four models instruction-tuned to identify harmfulness in both prompts and responses: Llama-Guard [16], Llama-Guard2 [26], Aegis-Guard-Defensive [13], and Aegis-Guard-Permissive [13]. For these models, to label refusal we mark a response as compliance if the output is harmful, and as a refusal if the output label is safe. We show results for harmful prompts only (RR(h) in Table 2), as this mapping is more likely to succeed in this setting. We also test LibrAI-LongFormer-ref, which is trained to detect refusals on the Do-Not-Answer benchmark [38] and used for evaluations in TrustLLM [32]. Lastly, we test a keyword-based classifier, which identifies refusals using a predefined list of refusal keywords (see our list of keywords in Appendix D.3). We evaluate the latter two models only on refusal detection. We also assess two API-based tools: GPT-4[1] [1] and OpenAI Moderation API [28]. For GPT-4 we experiment with various prompts and select the one that results in best performance. We provide this prompt in Table 15 of Appendix D.2. OpenAI Moderation API returns labels for toxicity, which we use for prompt and response harmfulness.

[1]We use `gpt-4-0125-preview` for all our experiments.

Finding 1: Existing open tools are unreliable on adversarial prompts and far behind GPT-4. We see in Figure 2 that existing open tools perform decently in detecting harm in vanilla prompts, but struggle to do so in adversarial prompts. For both prompt types we also see a sizeable performance gap for open tools relative to GPT-4, thus perpetuating costly reliance on API tools.

Finding 2: Existing open tools struggle with measuring refusals in model responses. Table 2 shows that open tools also struggle to identify refusals in models' completions. The top-performing harm-detection model, Llama-Guard2, trails GPT-4 by 15.1%, validating that response harmfulness is inadequate for this task—but even LibrAI-LongFormer-ref, a model trained specifically for refusal detection, only modestly outperforms the simple keyword-based baseline, and trails GPT-4 by 24.3%.

In summary, these patterns show non-trivial room for improvement in existing open-source tools, and a collection of related needs that can be filled by a high-quality multi-purpose moderation classifier.

3 Building WILDGUARDMIX and WILDGUARD

Motivated by the insights from §2 and by a lack of transparency in training of existing moderation tools, we develop two datasets, WILDGUARDTRAIN and WILDGUARDTEST as part of WILDGUARDMIX. Here we describe how we curate these datasets, and how we train WILDGUARD.

3.1 WILDGUARDTRAIN: Multi-task Moderation Training Dataset

WILDGUARDTRAIN (WGTRAIN) is a comprehensive training dataset of total 86,759 items, composed of data from diverse sources and amounting to 48,783 standalone prompts + 37,976 prompt-response pairs. The dataset covers both vanilla (direct request) and adversarial prompts across benign and harmful scenarios, as well as diverse types of refusals and compliances. WGTRAIN includes four sources of data—synthetic adversarial, synthetic vanilla, real-world user-LLM interactions (IN-THE-WILD), and existing annotator-written data— all selected to optimize coverage, diversity, and balance for our three tasks. Figure 1 provides an overview of the dataset composition.

3.1.1 Prompt Construction

Vanilla harmful synthetic prompts. We synthetically generate harmful prompts covering a broad range of risk scenarios. Inspired by Weidinger et al. [39], we consider 13 harm subcategories in four high-level categories: *privacy*, *misinformation*, *harmful language*, and *malicious uses*. Table 10 in Appendix A.6 lists harm categories with statistics of our data. To generate targeted, realistic, and diverse harmful scenarios, we design a structured generation pipeline, detailed in Appendix A.1.

Vanilla benign synthetic prompts. Similarly to WILDJAILBREAK [20], to enable maximally precise detection of prompt harmfulness, we additionally include two types of benign contrastive prompts: 1) benign prompts that superficially resemble unsafe prompts, motivated by the 10 exaggerated categories[2] from XSTest [30], and 2) benign prompts discussing sensitive but safe topics. These vanilla benign contrastive prompts are generated with GPT-4. Refer to Table 9 in Appendix A.2 for the GPT-4 prompt used to generate the queries and to Table 7 for examples.

Adversarial benign and harmful synthetic prompts. We employ the WILDTEAMING [20] framework to convert our generated vanilla prompts—harmful and benign—to adversarial prompts. This framework mines jailbreaking tactics from in-the-wild user-LLM interactions, and composes selections of tactics to transform vanilla requests into diverse adversarial counterparts. Specifically, WILDTEAMING first uses the OpenAI Moderation API to find potentially harmful user queries from LMSYS-CHAT-1M and WILDCHAT. Then we use prompted GPT-4 to decompose intent and jailbreak tactics from the queries. This mining process ends up with a pool for the jailbreak tactics; then we randomly sample 2-7 IN-THE-WILD tactics from the pool, and finally use the sampled tactics to transform vanilla prompts into adversarial prompts.

Prompts from IN-THE-WILD and existing annotator-written data. We consider two types of public data: 1) in-the-wild data that we label for harmfulness and 2) existing annotator-written safety

[2]Homonyms, figurative language, safe targets, safe contexts, definitions, real discrimination/nonsense group, nonsense discrimination/real group, historical events, public privacy, and fictional privacy.